

## Regular/Supplementary Winter Examination – 2024

**Subject Code & Name: Computer Architecture and Operation Systems (BTESC304\_AIDS)**

**Duration: 3 Hr.**

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

Page 1

|    |                                                                                                                                                                                                                                                                                               |        |   |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|---|
|    | the paging unit uses is :-<br>a) Linear conversion mechanism<br>b) One level table mechanism<br>c) Physical conversion mechanism<br>d) Two level table mechanism                                                                                                                              |        |   |
| 7  | What does MAR stand for?<br>a) Main Address Register      b) Memory Access Register<br>c) Main Accessible Register      d) Memory Address Register                                                                                                                                            | L1/CO3 | 1 |
| 8  | In memory-mapped I/O _____<br>a) The I/O devices and the memory share the same address space<br>b) The I/O devices have a separate address space<br>c) The memory and I/O devices have an associated address space<br>d) A part of the memory is specifically set aside for the I/O operation | L1/CO4 | 1 |
| 9  | What is an operating system?<br>a) Interface between the hardware and application programs<br>b) Collection of programs that manages hardware resources<br>c) System service provider to the application programs<br>d) All of the mentioned                                                  | L1/CO4 | 1 |
| 10 | The interval from the time of submission of a process to the time of completion is termed as _____<br>a) Waiting time<br>b) Turnaround time<br>c) Response time<br>d) Throughput                                                                                                              | L2/CO2 | 1 |
| 11 | When several processes access the same data concurrently and the outcome of the execution depends on the particular order in which the access takes place is called _____<br>a) Dynamic condition      b) Race condition<br>c) Essential condition      d) Critical condition                 | L2/CO4 | 1 |
| 12 | A deadlock avoidance algorithm dynamically examines the _____ to ensure that a circular wait condition can never exist.<br>a) Resource allocation state<br>b) System storage state<br>c) Operating system<br>d) Resources                                                                     | L2/CO4 | 1 |
|    |                                                                                                                                                                                                                                                                                               |        |   |

|            |                                                                                                                                                                                                                                                     |            |              |            |    |   |   |    |   |   |    |   |   |    |   |   |  |  |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|--------------|------------|----|---|---|----|---|---|----|---|---|----|---|---|--|--|
| Q. 2       | Solve the following.                                                                                                                                                                                                                                |            | 12           |            |    |   |   |    |   |   |    |   |   |    |   |   |  |  |
| A)         | Difference between Computer Architecture and Computer Organization.                                                                                                                                                                                 | L4/CO1     | 6            |            |    |   |   |    |   |   |    |   |   |    |   |   |  |  |
| B)         | Explain RISC and CISC architecture with net diagram.                                                                                                                                                                                                | L2/CO1     | 6            |            |    |   |   |    |   |   |    |   |   |    |   |   |  |  |
|            |                                                                                                                                                                                                                                                     |            |              |            |    |   |   |    |   |   |    |   |   |    |   |   |  |  |
| Q.3        | Solve the following.                                                                                                                                                                                                                                |            | 12           |            |    |   |   |    |   |   |    |   |   |    |   |   |  |  |
| A)         | List and Explain RAID levels with diagram.                                                                                                                                                                                                          | L1/CO2     | 6            |            |    |   |   |    |   |   |    |   |   |    |   |   |  |  |
| B)         | Describe Optimal page replacement algorithm and consider the page references 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 3 with 4-page frame. Find number of page fault using Optimal Page Replacement Algorithm.                                        | L3/CO2     | 6            |            |    |   |   |    |   |   |    |   |   |    |   |   |  |  |
|            |                                                                                                                                                                                                                                                     |            |              |            |    |   |   |    |   |   |    |   |   |    |   |   |  |  |
| Q. 4       | Solve Any Two of the following.                                                                                                                                                                                                                     |            | 12           |            |    |   |   |    |   |   |    |   |   |    |   |   |  |  |
| A)         | Describe Microprogrammed control unit with diagram.                                                                                                                                                                                                 | L2/CO3     | 6            |            |    |   |   |    |   |   |    |   |   |    |   |   |  |  |
| B)         | Explain the interrupt process from Interrupt driven I/O.                                                                                                                                                                                            | L2/CO3     | 6            |            |    |   |   |    |   |   |    |   |   |    |   |   |  |  |
| C)         | Define micro-operations of control unit.                                                                                                                                                                                                            | L2/CO3     | 6            |            |    |   |   |    |   |   |    |   |   |    |   |   |  |  |
|            |                                                                                                                                                                                                                                                     |            |              |            |    |   |   |    |   |   |    |   |   |    |   |   |  |  |
| Q.5        | Solve Any Two of the following.                                                                                                                                                                                                                     |            | 12           |            |    |   |   |    |   |   |    |   |   |    |   |   |  |  |
| A)         | Explain Objectives and Functions of Operating system.                                                                                                                                                                                               | L2/CO4     | 6            |            |    |   |   |    |   |   |    |   |   |    |   |   |  |  |
| B)         | List the different types of operations that can be performed on a process.                                                                                                                                                                          | L1/CO4     | 6            |            |    |   |   |    |   |   |    |   |   |    |   |   |  |  |
| C)         | What are the average waiting and turnaround times for the round-robin scheduling algorithm (RR) with a time quantum of 2 units?                                                                                                                     | L3/CO4     | 6            |            |    |   |   |    |   |   |    |   |   |    |   |   |  |  |
|            | <table><tr><td>Process ID</td><td>Arrival Time</td><td>Brust Time</td></tr><tr><td>P1</td><td>0</td><td>5</td></tr><tr><td>P2</td><td>1</td><td>4</td></tr><tr><td>P3</td><td>2</td><td>2</td></tr><tr><td>P4</td><td>4</td><td>1</td></tr></table> | Process ID | Arrival Time | Brust Time | P1 | 0 | 5 | P2 | 1 | 4 | P3 | 2 | 2 | P4 | 4 | 1 |  |  |
| Process ID | Arrival Time                                                                                                                                                                                                                                        | Brust Time |              |            |    |   |   |    |   |   |    |   |   |    |   |   |  |  |
| P1         | 0                                                                                                                                                                                                                                                   | 5          |              |            |    |   |   |    |   |   |    |   |   |    |   |   |  |  |
| P2         | 1                                                                                                                                                                                                                                                   | 4          |              |            |    |   |   |    |   |   |    |   |   |    |   |   |  |  |
| P3         | 2                                                                                                                                                                                                                                                   | 2          |              |            |    |   |   |    |   |   |    |   |   |    |   |   |  |  |
| P4         | 4                                                                                                                                                                                                                                                   | 1          |              |            |    |   |   |    |   |   |    |   |   |    |   |   |  |  |
| Q. 6       | Solve Any Two of the following.                                                                                                                                                                                                                     |            | 12           |            |    |   |   |    |   |   |    |   |   |    |   |   |  |  |
| A)         | Explain Critical section problem with its solution.                                                                                                                                                                                                 | L2/CO5     | 6            |            |    |   |   |    |   |   |    |   |   |    |   |   |  |  |
| B)         | Explain Thread in Operating Systems with its types.                                                                                                                                                                                                 | L2/CO5     | 6            |            |    |   |   |    |   |   |    |   |   |    |   |   |  |  |
| C)         | Define Deadlock? explain the four methods for handling deadlocks.                                                                                                                                                                                   | L2/CO5     | 6            |            |    |   |   |    |   |   |    |   |   |    |   |   |  |  |
|            | *** End ***                                                                                                                                                                                                                                         |            |              |            |    |   |   |    |   |   |    |   |   |    |   |   |  |  |